# Ahmad Raza

Bhakkar, Punjab, Pakistan
ahmadrazacdx.me
ahmedrazacdx@gmail.com
github.com/ahmadrazacdx
linkedin.com/in/ahmadrazacdx

## Research Interests

Representation learning, self-supervised learning for language, latent predictive learning, probabilistic generative models, generalization theory, and optimization dynamics.

## Education

2022 – 2026 **Thal University Bhakkar,** B.S. in Software Engineering

- Program Topper (Ranked 1$^{st}$ out of 80+ students) with CGPA: 3.86 / 4.0
- Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Statistics & Probability, Software Engineering, Database Systems, Cloud Computing

## Research

2026 **Ahmad Raza**. "CLAG: Stateless Cognitive Adaptation for RAG." In Preparation

- Proposed a stateless Bayesian framework that estimates a learner's cognitive level from a single query via posterior fusion with EM-estimated population priors.
- Derived and empirically validated theoretical guarantees, confirming single-query adaptation viability and strict adaptation monotonicity.
- Demonstrated via a comprehensive evaluation pipeline that the Bayesian formulation achieves highly accurate Zone of Proximal Development (ZPD) targeting (MAE $\approx$ 0.068) and significantly outperforms linear baselines ($p < 0.001$).

## Projects

Feb – May 2026 **Sage (FYP)** | LangChain, LangGraph, llama.cpp, FastAPI, React |

- Architected a 7-agent LangGraph system with dual-model inference and a hybrid RAG pipeline combining dense retrieval with BM25.
- Engineered a hardware-aware llama.cpp lifecycle manager with dynamic context-window scaling (3 K–64 K), VRAM-based GPU offloading, and cold-start reduction from 17 s to 4 s via Bayesian optimization (Optuna)-tuned compiler flags.
- Shipped as a Windows desktop application across four tiered builds (CPU/CUDA × full/lite) with SHA-256 integrity validation; CI/CD via GitHub Actions uploading to Cloudflare R2.
- Facilitated zero-configuration deployment in privacy-restricted environments via a schema-validated declarative configuration system (TOML) supporting fully offline execution in resource-constrained environments.

Nov 2025 **Sequence Modeling from Scratch** | Python, NumPy, Math |

- Implemented recurrent architectures (RNN, LSTM, GRU) from first principles (without auto-differentiation frameworks) to model gating mechanisms and computational graphs.
- Manually derived and coded analytical gradient computation via Backpropagation Through Time (BPTT) to manage gradient flow and solve vanishing/exploding gradients.
- Constructed sequence-to-sequence (Seq2Seq) models integrated with Bahdanau and Luong attention mechanisms, optimizing character and word-level language models.

Sep 2025 **NeuroScope** | Python, NumPy, Matplotlib |

- Developed an open-source diagnostic framework monitoring ten real-time neural network health indicators (e.g., dead neurons, gradient SNR, weight update ratios), validated against established initialization and optimization methods (Glorot, He, Pascanu).
- Designed interactive visualizations for activation distributions, gradient flow, and parameter weight histograms to analyze training dynamics and detect distribution shifts.
- Maintained extensive test coverage ($\geq$60%) with a fully automated packaging and distribution pipeline to PyPI via GitHub Actions.

## Awards

2020 – 2025 Punjab Educational Endowment Fund (PEEF) Scholarship, Government of Punjab. Competitive merit scholarship sustained over six consecutive years, from high school through undergraduate completion.

## Certifications

| | | |
|---|---|---|
| 2024 – 2025 | Deep Learning Specialization | DeepLearning.AI |
| 2024 – 2025 | Machine Learning Specialization | DeepLearning.AI |
| 2024 – 2025 | Mathematics for ML & Data Science Specialization | DeepLearning.AI |
| 2025 | MLOps Specialization | Duke University |
| 2025 | Machine Learning Professional Certificate | IBM |

## Skills

| | |
|---|---|
| Theoretical Foundations | Probability Theory, Statistical Learning, Optimization, Information Theory, Causal Inference |
| Programming Languages | Python, SQL |
| Libraries | NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, XGBoost, PyTorch, Transformers, LangChain, LangGraph, FastAPI |
| MLOps | llama.cpp, Weights & Biases, MLflow, DVC, Optuna |
| DevOps | Docker, Git, GitHub Actions, AWS |
| Languages | Urdu (Native), English (Conversational) |